

Problem Sheet 4

1. Let $\{Y_t\}$ be a process that satisfies the equation

$$(1 - B)^2 Y_t = (1 - \theta B) \varepsilon_t, \quad -\infty < t < \infty,$$

where B is the backward shift operator and $\{\varepsilon_t\}$ is a white noise process. Suppose that the process $\{Y_t\}$ has been observed up to time T and consider the minimum mean square error forecast for lead time $h \geq 1$, where H_T represents the history of the process up to time T :

$$\hat{y}_T(h) = E(Y_{T+h} | H_T).$$

- (a) State to which member of the family of ARIMA processes $\{Y_t\}$ belongs.
- (b) Expand the model equation to obtain an explicit expression for Y_t in terms of previous process values and white noise terms.
- (c) By setting $t = T + 1$ in the model equation, find an expression for $\hat{y}_T(1)$.
- (d) Write down a simple expression for the forecast error as a white noise term:

$$Y_{T+1} - \hat{y}_T(1).$$

- (e) By setting $t = T + 2$ in the model equation, show that

$$\hat{y}_T(2) = 3Y_T - 2Y_{T-1} - 2\theta\varepsilon_T.$$

- (f) Show that, for $h \geq 3$,

$$\hat{y}_T(h) = 2\hat{y}_T(h-1) - \hat{y}_T(h-2).$$

- (g) Deduce that

$$\hat{y}_T(h) = Y_T + b_T h, \quad h \geq 1,$$

where

$$b_T = Y_T - Y_{T-1} - \theta\varepsilon_T.$$

- (h) Show that the sequence $\{b_T\}$ satisfies the recurrence relation:

$$b_T = (1 - \theta)(Y_T - Y_{T-1}) + \theta b_{T-1}.$$

2. A data set of durations until payment for 130 consecutive orders from a goods distributor are in the `days` object in the `TSA` package. Type `library(TSA); data(days); print(days)` in R to see the data set.

- (a) Fit an MA(2) model and write the code to estimate the parameters using the Method of Moments. Compare the results with estimates using the least squares and the maximum likelihood methods.

```
# Fit MA(2) model using Least Squares
days.ma2.fit <- arima(days, order=c(0,0,1), method='CSS') ### Use 'ML' for maximum likelihood
days.ma2.fit
```

- (b) Forecast the next 10 values of the series, and list the forecasted values. Also plot the series along with the forecasted values and 95% prediction limits for the next ten values of the series.

Use the code below

```

n=length(days)
# Obtain MMSE forecasts
days.ma2.predict <- predict(days.ma2.fit,n.ahead=10)
round(days.ma2.predict$pred,3)
round(days.ma2.predict$se,3)
# Create lower and upper prediction interval bounds
lower.pi<-days.ma2.predict$pred-qnorm(0.975,0,1)*days.ma2.predict$se
upper.pi<-days.ma2.predict$pred+qnorm(0.975,0,1)*days.ma2.predict$se
# Display prediction intervals (2007-2026)
n1=n+1
n10=n+10
data.frame(Year=c(n1:n10),lower.pi,upper.pi)
# Note: Argument pch=16 produces a small black circle (for MMSE forecasts)
plot(days.ma2.fit,n.ahead=10,col='red',type='b',pch=16,ylab="Deviations (in inch)",xlab="Time")
# Put horizontal line at ML estimate of overall mean
abline(h=coef(days.ma2.fit)[names(coef(days.ma2.fit))=='intercept'])

```

- (c) There are three clear outliers (at times 63, 106, and 129) which can be seen from the time series plot. Replace these outliers with a “typical duration” value of 35. To do this, you can use the code

```

days.adj=days
days.adj[c(63,106,129)]=35
print(days.adj)

```

Fit an MA(2) model to this adjusted series using the three methods of estimation and compare the results.

- (d) Forecast the next 10 values of the adjusted series, and list the forecasted values. Also plot the series along with the forecasted values and 95% prediction limits for the next ten values of the series.
- (e) Comment on any differences between the forecasts based on the original data and the forecasts based on the adjusted data.

3. For the seasonal model:

$$Y_t = \Phi Y_{t-4} + e_t - \theta e_{t-1}, \quad \text{with } |\Phi| < 1, \quad (1)$$

find γ_0 and ρ_k .

4. Identify the following as certain multiplicative seasonal ARIMA models:

- (a) $Y_t = 0.5Y_{t-1} + Y_{t-4} - 0.5Y_{t-5} + e_t - 0.3e_{t-1}$.
- (b) $Y_t = Y_{t-1} + Y_{t-12} - Y_{t-13} + e_t - 0.5e_{t-1} - e_{t-12} + 0.5e_{t-13}$.

5. The quarterly earnings per share for 1960-1980 are in the JJ object in the TSA package. To load and print the dataset in R, use:

```

library(TSA)
data(JJ)
print(JJ)

```

- (a) Plot the original time series and the (natural) logarithm of the series. Explain why the log transformation is useful when modeling this series.
- (b) Explain why the log-transformed series is clearly not stationary. Plot the first differences of the log-transformed series. Could the differenced logged series be viewed as stationary?
- (c) Graph the sample ACF of the differenced logged data. What does this plot suggest?
- (d) Take both the first differences and the (lag-4) seasonal differences of the logged data. Plot this differenced and seasonally differenced series. Interpret what the plot indicates.
- (e) Plot the sample ACF of the differenced and seasonally differenced logged series. Interpret what the plot tells you.
- (f) Fit an $\text{ARIMA}(0,1,1) \times (0,1,1)_4$ model to the logged series, and assess the significance of the estimated coefficients.
- (g) Perform model diagnostics, based on the residuals.
- (h) Calculate the forecasts for the next 8 quarters of the series. Plot the forecasts along with 95% prediction intervals.